

Chapter 9 Roof Assemblies: 2023 FBC - Residential, 8th edition | UpCodes

Chapter 9 Roof Assemblies

Section R901 General

R901.1 Scope

The provisions of this chapter shall govern the design, materials, construction and quality of roof assemblies.

Exception: Buildings and structures located within the High-Velocity Hurricane Zone shall comply with the provisions of Chapter 44.

Section R902 Fire Classification

R902.1 Roofing Covering Materials

Roofs shall be covered with materials as set forth in Sections R904 and R905. Class A, B or C *roof assemblies* shall be installed in jurisdictions designated by law as requiring their use or where the edge of the roof is less than 3 feet (914 mm) from a lot line. Where Class A, B, or C *roof assemblies* are required, they shall be tested in accordance with ASTM E108 or UL 790. Where required, the *roof assembly* shall be *listed* and identified as to Class by an *approved* testing agency.

Exceptions:

1. Class A roof assemblies include those with coverings of brick, masonry and exposed concrete roof deck.
2. Class A roof assemblies include ferrous or copper shingles or sheets, metal sheets and shingles, clay or concrete roof tile, or slate installed on noncombustible decks.
3. Class A roof assemblies include minimum 16 ounces per square foot copper sheets installed over combustible decks.

R902.2 Fire-Retardant-Treated Shingles and Shakes

Fire-retardant-treated wood shakes and shingles shall be treated by impregnation with chemicals by the full-cell vacuum-pressure process, in accordance with AWPA C1. Each bundle shall be marked to identify the manufactured unit and the manufacturer, and shall be *labeled* to identify the classification of the material in accordance with the testing required in Section R902.1, the treating company and the quality control agency.

R902.3 Building-Integrated Photovoltaic Product

Reserved.

R902.4 Rooftop-Mounted Photovoltaic Panel Systems

Rooftop-mounted *photovoltaic panel systems* installed on or above the roof covering shall be tested, listed and identified with a fire classification in accordance with UL 1703 and UL 2703. Class A, B or C *photovoltaic panel systems* and modules shall be installed in jurisdictions designated by law as requiring their use or where the edge of the roof is less than 3 feet (914 mm) from a lot line.

Section R903 Weather Protection

R903.1 General

Roof decks shall be covered with *approved* roof coverings secured to the building or structure in accordance with the provisions of this chapter. Roof assemblies shall be designed and installed in accordance with this code and the *approved* manufacturer's installation instructions such that the roof assembly shall serve to protect the building or structure.

R903.2 Flashing

Flashings shall be used to seal roofing systems, where the system is interrupted or terminated and shall be installed in a manner that prevents moisture from entering the wall and roof through joints in copings, through moisture permeable materials and at

intersections with parapet walls and other penetrations through the roof plane.

R903.2.1 Locations

Flashings shall be installed at wall and roof intersections, wherever there is a change in roof slope or direction and around roof openings. Where flashing is of metal, the metal shall be corrosion resistant with a thickness of not less than provided in Table R903.2.1 or in compliance with RAS 111.

Exception: Flashing is not required at hip and ridge junctions.

TABLE R903.2.1

METAL FLASHING MATERIAL

MATERIAL	GAGE MINIMUM THICKNESS (inches)	GAGE	WEIGHT (lbf)
Copper	0.024	—	1 (16 oz)
Aluminum	0.024	—	—
Stainless steel	—	28	—
Galvanized steel	0.0179	26 (zinc coated G90)	26 (zinc coated G90)
Aluminum zinc coated steel	0.0179	26 (AZ50 alum zinc)	26 (AZ50 alum zinc)
Zinc alloy	0.027	—	—
Lead	—	2.5 (40 oz.)	—
Painted terne	—	—	1.25 (20 oz.)

R903.2.2 Crickets and Saddles

A cricket or saddle shall be installed on the ridge side of any chimney or penetration more than 30 inches (762 mm) wide as measured perpendicular to the slope. Cricket or saddle coverings shall be sheet metal or of the same material as the roof covering.

Exception: Unit skylights installed in accordance with Section R308.6 and flashed in accordance with the manufacturer's instructions shall be permitted to be installed without a cricket or saddle.

R903.2.3 Membrane Flashings

All membrane flashing shall be installed according to the roof assembly manufacturer's published literature.

R903.3 Coping

Parapet walls shall be properly coped with noncombustible, weatherproof materials of a width not less than the thickness of the parapet wall.

R903.4 Roof Drainage

Unless roofs are sloped to drain over roof edges, roof drains shall be installed at each low point of the roof. Where required for roof drainage, scuppers shall be placed level with the roof surface in a wall or parapet. The scupper shall be located as determined by the roof slope and contributing roof area.

R903.4.1 Overflow Drains and Scuppers

When other means of drainage of overflow water is not provided, overflow scuppers shall be placed in walls or parapets not less than 2 inches (51 mm) nor more than 4 inches (102 mm) above the finished roof covering and shall be located as close as practical to required vertical leaders or down-spouts or wall and parapet scuppers. An overflow scupper shall be sized in accordance with the *Florida Building Code, Plumbing*. Overflow drains shall discharge to an *approved* location and shall not be connected to roof drain lines.

R903.4.2 One and Two Family Dwellings, and Private Garages

When gutters and leaders are placed on the outside of buildings, the gutters and leaders shall be constructed of metal or approved plastic for outdoor exposure with lapped, soldered or caulked joints and shall be securely fastened to the building with a corrosion resistant fastening device of similar or compatible material to the gutters and downspouts.

Section R904 Materials

R904.1 Scope

The requirements set forth in this section shall apply to the application of roof covering materials specified herein. Roof assemblies shall be applied in accordance with this chapter and the manufacturer's installation instructions. Installation of roof assemblies shall comply with the applicable provisions of Section R905.

R904.2 Compatibility of Materials

Roof assemblies shall be of materials that are compatible with each other and with the building or structure to which the materials are applied.

R904.3 Material Specifications and Physical Characteristics

Roof covering materials shall conform to the applicable standards listed in this chapter. In the absence of applicable standards or where materials are of questionable suitability, testing by an *approved* testing agency shall be required by the *building official* to determine the character, quality and limitations of application of the materials.

R904.4 Product Identification

Roof covering materials shall be delivered in packages bearing the manufacturer's identifying marks and *approved* testing agency *labels* required. Bulk shipments and/or site manufactured materials shall be accompanied by the same information or issued in the form of a certificate or on a bill of lading by the manufacturer.

R904.5 Fasteners

R904.5.1 Nails

Nails shall be corrosion-resistant nails conforming to ASTM F1667 or an equal corrosion resistance by coating, electro galvanization, mechanical galvanization, hot dipped galvanization, stainless steel, nonferrous metal and alloys or other suitable corrosion-resistant material, or corrosion resistance shall be demonstrated in accordance with TAS 114, Appendix E.

R904.5.2 Screws

Wood screws shall conform to ANSI/ASME B 18.6.1. Screws shall be corrosion resistant by coating, galvanization, stainless steel, nonferrous metal or other suitable corrosion-resistant material. The corrosion resistance shall be demonstrated through one of the following methods:

1. Corrosion resistance equivalent to ASTM A641, Class 1;
2. Corrosion resistance in accordance with TAS 114, Appendix E;
3. Corrosion-resistant coating exhibiting not more than 5-percent red rust after 1000 hours exposure in accordance with ASTM B117.

R904.5.3 Clips

Clips shall be corrosion-resistant clips. The corrosion resistance shall meet 0.90 ounce per square foot (0.458 kg/m²) measured according ASTM A90/A90M, TAS 114 Appendix E or an equal corrosion resistance coating, electro galvanization, mechanical

galvanization, hot dipped galvanization, stainless steel, nonferrous metals and alloys or other suitable corrosion-resistant material. Stainless steel clips shall conform to ASTM A240/A240M, Type 304.

Section R905 Requirements for Roof Coverings

R905.1 Roof Covering Application

Roof coverings shall be applied in accordance with the applicable provisions of this section and the manufacturer's installation instructions. Unless otherwise specified in this section, roof coverings shall be installed to resist the component and cladding loads specified in Table R301.2(2), adjusted for height and exposure in accordance with Table R301.2(3).

R905.1.1 Underlayment

Underlayment for roof slopes 2:12 and greater shall conform to the applicable standards listed in this chapter. Underlayment materials required to comply with ASTM D226, D1970, D4869, D6757, or ASTM D8257, shall bear a label indicating compliance to the standard designation and, if applicable, type classification indicated. Underlayment for roof slopes 2:12 and greater shall be applied and attached in accordance with Section R905.1.1.1 or R905.1.1.2 as applicable.

Exception: Compliance with Section R905.1.1.1 is not required for structural metal panels that do not require a substrate or underlayment.

Table R905.1.1(1) Underlayment Types. Reserved.

Table R905.1.1(2) Underlayment Application. Reserved.

Table R905.1.1(3) Underlayment Attachment. Reserved.

R905.1.1.1 Underlayment for Asphalt Shingles, Metal Roof Panels or Shingles, Mineral Surfaced Roll Roofing, Slate and Slate-Type Shingles, Wood Shakes and Wood Shingles

Underlayment for asphalt shingles, metal roof panels or shingles, mineral surfaced roll roofing, slate and slate-type shingles, wood shakes and wood shingles shall comply with one of the following methods:

1. The entire roof deck shall be covered with an approved self-adhering polymer-modified bitumen underlayment complying with ASTM D1970 installed in accordance with both the underlayment manufacturer's and roof covering manufacturer's installation instructions for the deck material, roof ventilation configuration and climate exposure for the roof covering to be installed.

Exceptions:

1. This method is not permitted for wood shingles or shakes.
2. An existing self-adhering modified bitumen underlayment that has been previously installed over the roof decking and, where it is required, reroofing off the roof sheathing in accordance with Section 706.7.1 of the *Florida Building Code, Existing Building* can be confirmed or verified. An approved underlayment in accordance with Table R905.1.1.1 for the applicable roof covering shall be applied over the entire roof over the existing self-adhered modified bitumen underlayment.
2. A minimum 3³/₄-inch-wide (95 mm) strip of self-adhering polymer-modified bitumen membrane complying with ASTM D1970 or self-adhering flexible flashing tape complying with AAMA 711, Level 3 [for exposure up to 176°F (80°C)], installed in accordance with the manufacturer's instructions for the deck material, shall be applied over all joints in the roof decking. An approved underlayment in accordance with Table R905.1.1.1 for the applicable roof covering shall be applied over the entire roof over the membrane strips.
3. Two layers of ASTM D226 Type II, ASTM D4869 Type III or Type IV or ASTM D8257 underlayment shall be installed as follows: Apply a strip of underlayment for the first course that is half the width of a full sheet parallel to and starting at the eaves, fastened sufficiently to hold in place. Starting at the eave, apply a full sheet of underlayment, for the second course. Apply the third course of underlayment overlapping the second course half the width of a full sheet plus 2 inches

(51 mm). Overlap all successive courses half the width of a full sheet plus 1 inch (25 mm). End laps shall be 6 inches (152 mm) and shall be offset by 6 feet (1829 mm). Underlayment shall be attached to a nailable deck with corrosion-resistant fasteners with a maximum fastener spacing, measured horizontally and vertically, of 12 inches (305 mm) o.c. between side laps, and one row at the end and side laps fastened 6 inches (152 mm) o.c. Underlayment shall be attached using annular ring or deformed shank nails with metal or plastic caps with a nominal cap diameter of not less than 1 inch (25 mm). Metal caps are required where the ultimate design wind speed, V_{ult} , equals or exceeds 170 mph. Metal caps shall have a thickness of not less than 32-gage sheet metal. The minimum thickness of the outside edge of plastic caps shall be 0.035 inch (0.889 mm). The cap nail shank shall be not less than 0.083 inch (2.1082 mm) for ring shank nails. The cap nail shank shall have a length sufficient to penetrate through the roof sheathing or not less than $\frac{3}{4}$ inch into the roof sheathing.

Exception: Use of ASTM D8257 underlayment is not permitted for wood shingles or shakes.

TABLE R905.1.1.1

UNDERLAYMENT WITH SELF-ADHERING STRIPS OVER ROOF DECKING JOINTS

ROOF COVERING	UNDERLAYMENT TYPE	UNDERLAYMENT ATTACHMENT	
		ROOF SLOPE 2:12 AND LESS THAN 4:12	ROOF SLOPE 4:12 AND GREATER
Asphalt Shingles, Metal Roof Panels, Photovoltaic Shingles	ASTM D226 Type II ASTM D4869 Type III or IV ASTM D6757, ASTM D8257	Apply in accordance with Section R905.1.1.1, Item 3.	Underlayment shall be applied shingle fashion, parallel to and starting from the eave and lapped 4 inches (51 mm); end laps shall be 6 inches and shall be offset by 6 feet. Underlayment shall be attached to a nailable deck with corrosion-resistant fasteners with a maximum fastener spacing measured horizontally and vertically of 12 inches (305 mm) o.c. between side laps and one row at the end and side laps fastened 6 inches (152 mm) o.c. Underlayment shall be attached using annular ring or deformed shank nails with metal or plastic caps with a nominal cap diameter of not less than 1 inch. Metal caps are required where the ultimate design wind speed, V_{ult} , equals or exceeds 170 mph. Metal caps shall have a thickness of not less than 32-gage sheet metal. Power-driven metal caps shall have a minimum thickness of 0.010 inch. Minimum thickness of the outside edge of plastic caps shall be 0.035 inch. The cap nail shank shall be not less than 0.083 inch for ring shank cap nails. Cap nail shank shall have a length sufficient to penetrate through the roof sheathing or not less than $\frac{3}{4}$ inch into the roof sheathing.
Metal Roof Shingles, Mineral-Surface Roll Roofing, Slate and Slate-type Shingles	ASTM D226 Type II ASTM D4869 Type III or IV ASTM D8257		
Wood Shingles, Wood Shakes	ASTM D226 Type II ASTM D4869 Type III or IV		

R905.1.1.2 Underlayment for Concrete and Clay Tile

Underlayment for concrete and clay tile shall comply with Section 905.3.3.

R905.1.2 Ice Barriers

Reserved.

R905.2 Asphalt Shingles

The installation of asphalt shingles shall comply with the provisions of this section or RAS 115.

R905.2.1 Sheathing Requirements

Asphalt shingles shall be fastened to solidly sheathed decks.

R905.2.2 Slope

Asphalt shingles shall be used only on roof slopes of two units vertical in 12 units horizontal (2:12) or greater.

R905.2.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.1.1.

R905.2.4 Asphalt Shingles

Asphalt shingles shall comply with ASTM D3462.

R905.2.4.1 Installation of Asphalt Shingles

Asphalt shingles shall be installed in accordance with Sections R905.2.6 and R905.2.6.1.

Table R905.2.4.1 Classification of Asphalt Roof Shingles. Reserved.

R905.2.5 Fasteners

Fasteners for asphalt shingles shall be galvanized steel, stainless steel, aluminum or copper roofing nails, minimum 12-gage [0.105 inch (3 mm)] shank with a minimum $\frac{3}{8}$ -inch-diameter (9.5 mm) head, complying with ASTM F1667, of a length to penetrate through the roofing materials and not less than $\frac{3}{4}$ inch (19.1 mm) into the roof sheathing. Where the roof sheathing is less than $\frac{3}{4}$ inch (19.1 mm) thick, the fasteners shall penetrate through the sheathing.

Exception: If the architectural appearance is to be preserved from below, an alternate method of attachment complying with the wind load requirements of Chapter 16 of the *Florida Building Code, Building* may be proposed unless otherwise addressed in Chapter 9. The alternative attachment shall be prepared, signed and sealed by a Florida-registered architect or a Florida-registered engineer, which architect or engineer shall be proficient in structural design.

R905.2.6 Attachment

Asphalt shingles shall have the minimum number of fasteners required by the manufacturer, but not less than four fasteners per strip shingle or two fasteners per individual shingle. Where the roof slope exceeds 21 units vertical in 12 units horizontal (21:12, 175-percent slope), shingles shall be installed as required by the manufacturer.

R905.2.6.1 Wind Resistance of Asphalt Shingles

Asphalt shingles shall be tested and classified in accordance with ASTM D3161, TAS 107 or ASTM D7158 and shall meet the required classification in accordance with Table R905.2.6.1 to resist the basic wind speed in accordance with Figure R301.2(4). Asphalt shingle wrappers shall be *labeled* to indicate compliance with one of the required classifications, as shown in Table R905.2.6.1.

TABLE R905.2.6.1

CLASSIFICATION OF ASPHALT SHINGLES

Maximum Basic Wind Speed, V_{ult} , From Figure R301.2(4)	V_{asd} as determined in accordance with Section R301.2.1.3	ASTM D7158	ASTM D3161
110	85	D, G or H	D or F
116	90	D, G or H	D or F
129	100	G or H	D or F
142	110	G or H	F
155	120	G or H	F
168	130	H	F
181	140	H	F
194	150	H	F

R905.2.7 Ice Barrier

Reserved.

R905.2.8 Flashing

Flashing for asphalt shingles shall comply with this section or RAS 111.

R905.2.8.1 Base and Counter Flashing

Base and counter flashing shall be installed as follows:

1. in accordance with manufacturer's installation instructions, or
2. in compliance with RAS 111, or
3. a continuous metal minimum 4 inch by 4 inch "L" flashing shall be set in approved flashing cement and set flush to base of wall and over the underlayment. Both horizontal and vertical metal flanges shall be fastened 6 inches (152 mm) on center with approved fasteners. All laps shall be a minimum of 4 inches (102 mm) fully sealed in approved flashing cement. Flashing shall start at the lower portion of roof to ensure water-shedding capabilities of all metal laps. The entire edge of the horizontal flange shall be sealed covering all nail penetrations with approved flashing cement and membrane. Shingles shall overlap the horizontal flange and shall be set in approved flashing cement.

Base flashing shall be of either corrosion-resistant metal provided in Section R905.2.8.1 or mineral surface roll roofing weighing a minimum of 77 pounds per 100 square feet (3.76 kg/m²). Counter flashing shall be corrosion-resistant metal with a minimum thickness provided in Table R903.2.1.

R905.2.8.2 Valleys

Valley linings shall be installed in accordance with the manufacturer's instructions before applying shingles. Valley linings of the following types shall be permitted:

1. For open valleys (valley lining exposed) lined with metal, the valley lining shall be not less than 16 inches (406 mm) wide and of any of the corrosion-resistant metals in Table R903.2.1.
2. For open valleys, valley lining of two plies of mineral-surfaced roll roofing, complying with ASTM D3909 or ASTM D6380 Class M, shall be permitted. The bottom layer shall be 18 inches (457 mm) and the top layer not less than 36 inches (914 mm) wide.
3. For closed valleys (valley covered with shingles), valley lining of one ply of smooth roll roofing complying with ASTM D6380 Class S and not less than 36 inches wide (914 mm) or valley lining as described in Item 1 or 2 shall be permitted. Self-adhering polymer modified bitumen underlayment complying with ASTM D1970 and not less than 36 inches (914 mm) wide shall be permitted in lieu of the lining material.

Table R905.2.8.2 Valley Lining Material. Reserved.

R905.2.8.3 Sidewall Flashing

Flashing against a vertical sidewall shall be by the step-flashing method or continuous "L" flashing method.

R905.2.8.4 Other Flashing

Flashing against a vertical front wall, as well as soil stack, vent pipe and chimney flashing, shall be applied in accordance with the asphalt shingle manufacturer's instructions.

R905.2.8.5 Drip Edge

Provide drip edge at eaves and gables of shingle roofs. Overlap to be a minimum of 3 inches (76 mm). Eave drip edges shall extend 1/2 inch (13 mm) below sheathing and extend back on the roof a minimum of 2 inches (51 mm). Drip edge shall be installed over the underlayment. Self-adhering ASTM D1970 underlayment may be installed over a primed drip edge flange. There shall be a minimum 4 inch (51 mm) width of roof cement installed over the drip edge flange or the self-adhering

underlayment. Drip edge shall be mechanically fastened a maximum of 12 inches (305 mm) on center. Where the V_{asd} as determined in accordance with Section R301.2.1.3 is 110 mph (177 km/h) or greater or the mean roof height exceeds 33 feet (10 058 mm), drip edges shall be mechanically fastened a maximum of 4 inches (102 mm) on center.

R905.3 Clay and Concrete Tile

The installation of clay and concrete tile shall be in accordance with the manufacturer's installation instructions, or recommendations of FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3.

R905.3.1 Deck Requirements

Concrete and clay tile shall be installed only over solid sheathing, except where the roof covering is specifically designed and tested in accordance with Chapter 16, *Florida Building Code, Building* to be applied over spaced structural sheathing boards.

R905.3.2 Deck Slope

Clay and concrete roof tile shall be installed on roof slopes in accordance with the recommendations of FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3.

R905.3.3 Underlayment

Unless otherwise noted, required underlayment shall comply with the underlayment manufacturer's installation instructions in accordance with the FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3.

Exception: Where an existing self-adhering modified bitumen underlayment that has been previously installed over the roof decking and, where it is required, renailling of the roof sheathing in accordance with Section 706.7.1 of the *Florida Building Code, Existing Building* can be confirmed or verified. An approved underlayment in accordance with a two-ply system as described in the FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition shall be applied over the entire roof over the existing self-adhered modified bitumen underlayment.

R905.3.3.1 Slope and Underlayment Requirements

Refer to manufacturer's installation instructions, FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3 for underlayment and slope requirements for specific roof tile systems.

R905.3.4 Clay Tile

Clay roof tile shall comply with ASTM C1167.

R905.3.5 Concrete Tile

Concrete roof tile shall comply with ASTM C1492.

R905.3.6 Fasteners

Nails shall be corrosion resistant and not less than 11 gage, $\frac{5}{16}$ -inch (11 mm) head, and of sufficient length to penetrate the deck not less than $\frac{3}{4}$ inch (19 mm) or through the thickness of the deck, whichever is less or in accordance with the FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3. Attaching wire for clay or concrete tile shall not be smaller than 0.083 inch (2.1 mm).

R905.3.7 Application

Tile shall be applied in accordance with this chapter and the manufacturer's installation instructions, recommendations of the FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition.

Table R905.3.7 Clay and Concrete Tile Attachment. Reserved.

R905.3.7.1 Hip and Ridge Tiles

Hip and ridge tiles shall be installed in accordance with FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asd} is determined in accordance with Section R301.2.1.3.

R905.3.8 Flashing

At the juncture of roof vertical surfaces, flashing and counterflashing shall be provided in accordance with this chapter and the manufacturer's installation instructions, recommendations of the FRSA/TRI *Florida High Wind Concrete and Clay Roof Tile Installation Manual*, Seventh Edition where the V_{asq} is determined in accordance with Section R301.2.1.3.

R905.4 Metal Roof Shingles

The installation of metal roof shingles shall comply with the provisions of this section. Metal roofing shingles shall be factory or field manufactured in accordance with the manufacturers' product approval specifications and limitations of use. Metal roofing shingles shall be factory or field manufactured under a quality assurance program that is audited by a third-party quality assurance entity approved by the Florida Building Commission for that purpose.

R905.4.1 Deck Requirements

Metal roof shingles shall be applied to a solid or closely fitted deck, except where the roof covering is specifically designed to be applied to spaced sheathing.

R905.4.2 Deck Slope

Metal roof shingles shall not be installed on roof slopes below three units vertical in 12 units horizontal (25-percent slope).

R905.4.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.11.

R905.4.3.1 Ice Barrier

Reserved.

R905.4.4 Material Standards

Metal roof shingle roof coverings shall comply with Table R905.4.4. The materials used for metal roof shingle roof coverings shall be naturally corrosion resistant or be made corrosion resistant in accordance with the standards and minimum thicknesses listed in Table R905.4.4.

TABLE R905.4.4

METAL ROOF COVERINGS

ROOF COVERING TYPE	STANDARD	STANDARD APPLICATION RATE/THICKNESS
Aluminum	ASTM B209	0.024-inch minimum thickness for roll-formed panels and 0.019-inch minimum thickness for press-formed shingles
Aluminum-zinc coated steel	ASTM A792	0.013-inch minimum thickness, AZ 50 (coated minimum application rate)
Cold-rolled copper	ASTM B370	Minimum 16 oz./sq. ft. and 12 oz./sq. ft. high yield copper for metal-sheet roof covering systems: 12 oz./sq. ft. for preformed metal shingle systems
Copper	ASTM B370	16 oz./sq. ft. for metal-sheet roof-covering systems; 12 oz./sq. ft. for preformed metal shingle systems.
Galvanized steel	ASTM A653	0.013-inch minimum thickness, G-90 zinc-coated ^a
Hard lead	—	2 lbs./sq. ft.
Lead-coated copper	ASTM B101	—

Prepainted steel	ASTM A755	0.0142 inch minimum thickness
Soft lead	—	3 lbs./sq. ft.
Stainless steel	ASTM A240	300 Series alloys
Steel	ASTM A924/ ASTM A924M	—
Terne and terne-coated stainless	—	Terne coating of 40 lbs. per double base box, field painted where applicable in accordance with manufacturer's installation instructions
Zinc	—	0.027 inch minimum thickness; 99.995% electrolytic high grade zinc with alloy additives of copper (0.08% — 0.20%), titanium (0.07% — 0.12%) and aluminum (0.015%)

For SI: 1 ounce per square foot = 0.0026 kg/m², 1 pound per square foot = 4.882 kg/m², 1 inch = 25.4 mm, 1 pound = 0.454 kg.

- a. For Group U buildings, the minimum coating thickness for ASTM A653 galvanized steel roofing shall be G-60.

R905.4.4.1 Wind Resistance of Metal Roof Shingles

Metal roof shingles applied to a solid or closely fitted deck shall be tested in accordance with ASTM D3161, FM 4474, UL 580, UL 1897 or TAS 107. Metal roof shingles tested in accordance with ASTM D3161 shall meet the classification requirements of Table R905.4.4.1 for the appropriate maximum basic wind speed and the metal shingle packaging shall bear a label to indicate compliance with ASTM D3161 and the required classification in Table R905.4.4.1.

TABLE R905.4.4.1

CLASSIFICATION OF METAL ROOF SHINGLES TESTED IN ACCORDANCE WITH ASTM D3161

MAXIMUM BASIC WIND SPEED FROM FIGURE R301.2(4) or ASCE 7	<i>V</i>_{asd}	ASTM D3161
110	85	D or F
116	90	D or F
129	100	D or F
142	110	F
155	120	F
168	130	F
181	140	F
194	150	F

R905.4.5 Application

Metal roof shingles shall be installed in accordance with the approved manufacturer's installation instructions. The product approval shall state the allowable uplift resistance for the attachment system. The installation of metal roof shingles shall be

limited to roofs where the allowable uplift resistance is equal to or greater than the design uplift pressure for the roof listed in Table R301.2(2).

R905.4.6 Flashing

Roof valley flashing shall be of corrosion-resistant metal of the same material as the roof covering or shall comply with the standards in Table R905.4.4. Valley flashing shall extend not less than 8 inches (203 mm) from the centerline each way and shall have a splash diverter rib not less than $\frac{3}{4}$ inch (19 mm) in height at the flow line formed as part of the flashing. Sections of flashing shall have an end lap of not less than 4 inches (102 mm). The metal valley flashing shall have a 36-inch-wide (914 mm) underlayment directly under it consisting of one layer of underlayment running the full length of the valley, in addition to underlayment required for metal roof shingles.

R905.5 Mineral-Surfaced Roll Roofing

The installation of mineral-surfaced roll roofing shall comply with this section.

R905.5.1 Deck Requirements

Mineral-surfaced roll roofing shall be fastened to solidly sheathed roofs.

R905.5.2 Deck Slope

Mineral-surfaced roll roofing shall not be applied on roof slopes below one unit vertical in 12 units horizontal (8-percent slope).

R905.5.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.1.1.

R905.5.3.1 Ice Barrier

Reserved.

R905.5.4 Material Standards

Mineral-surfaced roll roofing shall conform to ASTM D3909 or ASTM D6380, Class M or Class WS.

R905.5.5 Application

Mineral-surfaced roll roofing shall be installed in accordance with this chapter and the manufacturer's instructions.

R905.6 Slate and Slate-Type Shingles

The installation of slate and slate-type shingles shall comply with the provisions of this section.

R905.6.1 Deck Requirements

Slate shingles shall be fastened to solidly sheathed roofs.

R905.6.2 Deck Slope

Slate shingles shall be used only on slopes of four units vertical in 12 units horizontal (33-percent slope) or greater.

R905.6.3 Underlayment

Underlayment shall comply with Section R905.1.1. Underlayment shall be installed in accordance with the manufacturer's installation instructions.

R905.6.3.1 Ice Barrier

Reserved.

R905.6.4 Material Standards

Slate shingles shall comply with ASTM C406.

R905.6.5 Application

Minimum headlap for slate shingles shall be in accordance with Table R905.6.5. Slate shingles shall be secured to the roof with two fasteners per slate. Slate shingles shall be installed in accordance with this chapter and the manufacturer's instructions. The product approval shall state the allowable uplift resistance for the attachment system. The installation of slate and slate-type

shingles shall be limited to roofs where the allowable uplift resistance is equal to or greater than the design uplift pressure for the roof listed in Table R301.2(2).

TABLE R905.6.5

SLATE SHINGLE HEADLAP

SLOPE	HEADLAP (inches)
4:12 = slope < 8:12	4
8:12 = slope < 20:12	3
Slope = 20:12	2

For SI: 1 inch = 25.4 mm.

R905.6.6 Flashing

Flashing and counterflashing shall be made with sheet metal. Valley flashing shall be a minimum of 16 inches (406 mm) wide. Valley and flashing metal shall be a minimum thickness as provided in Table R903.2.1 for nonferrous metal or stainless steel. Chimneys, stucco or brick walls shall have not less than two plies of felt for a cap flashing consisting of a 4-inch-wide (102 mm) strip of felt set in plastic cement and extending 1 inch (25 mm) above the first felt and a top coating of plastic cement. The felt shall extend over the base flashing 2 inches (51 mm).

R905.6.7

Slate and slate-type shingles shall be installed in accordance with this chapter and the manufacturer's installation instructions. The product approval shall state the allowable uplift resistance for the attachment system. The installation of slate and slate-type shingles shall be limited to roofs where the allowable uplift resistance is equal to or greater than the design uplift pressure for the roof listed in Table R301.2(2).

R905.7 Wood Shingles

All wood shingles shall be installed in accordance with this chapter and the manufacturer's installation instructions, the Cedar Shake and Shingle Bureau *New Roof Construction Manual* or RAS 130. The product approval shall state the allowable uplift resistance for the attachment system. The installation of wood shingles shall be limited to roofs where the allowable uplift resistance is equal to or greater than the design uplift pressure for the roof listed in Table R301.2(2).

R905.7.1 Deck Requirements

Reserved.

R905.7.1.1 Solid Sheathing Required

Reserved.

R905.7.2 Deck Slope

Reserved.

R905.7.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.11.

R905.7.3.1 Ice Barrier

Reserved.

R905.7.4 Material Standards

Wood shingles shall be of naturally durable wood and comply with the requirements of Table R905.7.4.

TABLE R905.7.4

WOOD SHINGLE MATERIAL REQUIREMENTS

MATERIAL	MINIMUM GRADES	APPLICABLE GRADING RULES
Wood shingles of naturally durable wood	1, 2 or 3	Cedar Shake and Shingle Bureau

R905.7.5 Application

Reserved.

Table R905.7.5(1) Wood Shingle Weather Exposure and Roof Slope. Reserved.

Table R905.7.5(2) Nail Requirements for Wood Shakes and Wood Shingles. Reserved.

R905.7.6 Valley Flashing

Reserved.

R905.7.6.1 Fasteners

R905.7.6.1.1 Nails

Fasteners to attach wood shingles shall be Type 304 (Type 316 for coastal areas) stainless steel ring-shank nails with a minimum penetration of 0.75 inch into the sheathing. Each shingle shall be attached with a minimum of two fasteners.

R905.7.7 Label Required

Each bundle of shingles shall be identified by a *label* of an *approved* grading or inspection bureau or agency.

R905.8 Wood Shakes

All wood shakes shall be installed in accordance with this chapter and the manufacturer's installation instructions, the Cedar Shake and Shingle Bureau *New Roof Construction Manual* or RAS 130. The product approval shall state the allowable uplift resistance for the attachment system. The installation of wood shakes shall be limited to roofs where the allowable uplift resistance is equal to or greater than the design uplift pressure for the roof listed in Table R301.2(2).

R905.8.1 Deck Requirements

Reserved.

R905.8.1.1 Solid Sheathing Required

Reserved.

R905.8.2 Deck Slope

Reserved.

R905.8.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.11.

R905.8.3.1 Ice Barrier

Reserved.

R905.8.4 Interlayment

Reserved.

R905.8.5 Material Standards

Wood shakes shall comply with the requirements of Table R905.8.5.

TABLE R905.8.5

WOOD SHAKE MATERIAL REQUIREMENTS

MATERIAL	MINIMUM GRADES	APPLICABLE GRADING RULES
Wood shakes of naturally durable wood	1	Cedar Shake and Shingle Bureau
Tapersawn shakes of naturally durable wood	1 or 2	Cedar Shake and Shingle Bureau
Preservative-treated shakes and shingles of naturally durable wood	1	Cedar Shake and Shingle Bureau
Fire-retardant-treated shakes and shingles of naturally durable wood	1	Cedar Shake and Shingle Bureau
Preservative-treated tapersawn shakes of Southern pine treated in accordance with AWP Standard U1 (Commodity Specification A, Special Requirement 4.6)	1 or 2	Forest Products Laboratory of the Texas Forest Services

R905.8.6 Application

Reserved.

Table R905.8.6 Wood Shake Weather Exposure and Roof Slope. Reserved.

R905.8.7 Shake Placement

Reserved.

R905.8.7.1 Fasteners

R905.8.7.1.1 Nails

Fasteners to attach wood shakes shall be Type 304 (Type 316 for coastal areas) stainless steel ring-shank nails with a minimum penetration of 0.75 inch into the sheathing. Each shake shall be attached with a minimum of two fasteners.

R905.8.8 Valley Flashing

Reserved.

R905.8.9 Label Required

Each bundle of shakes shall be identified by a *label* of an *approved* grading or inspection bureau or agency.

R905.9 Built-Up Roofs

The installation of built-up roofs shall comply with the provisions of this section.

R905.9.1 Slope

Built-up roofs shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) for drainage, except for coal-tar built-up roofs, which shall have a design slope of a minimum one-eighth unit vertical in 12 units horizontal (1-percent slope).

R905.9.2 Material Standards

Built-up roof covering materials shall comply with the standards in Table R905.9.2 or UL 55A.

TABLE R905.9.2

BUILT-UP ROOFING MATERIAL STANDARDS

MATERIAL STANDARD	STANDARD
Acrylic coatings used in roofing	ASTM D6083
Aggregate surfacing	ASTM D1863
Asphalt adhesive used in roofing	ASTM D3747
Asphalt cements used in roofing	ASTM D2822; D3019; D4586
Asphalt-coated glass fiber base sheet	ASTM D4601
Asphalt coatings used in roofing	ASTM D 1227; D2823; D2824; D4479
Asphalt glass felt	ASTM D2178
Asphalt primer used in roofing	ASTM D41
Asphalt-saturated and asphalt-coated organic felt base sheet	ASTM D2626
Asphalt-saturated organic felt (perforated)	ASTM D226
Asphalt used in roofing	ASTM D312
Coal-tar cements used in roofing	ASTM D4022; D5643
Coal-tar primer used in roofing, dampproofing and waterproofing	ASTM D43
Coal-tar saturated organic felt	ASTM D227
Coal-tar used in roofing	ASTM D450, Type I or II
Glass mat, coal tar	ASTM D4990
Glass mat, venting type	ASTM D4897
Mineral-surfaced inorganic cap sheet	ASTM D3909
Thermoplastic fabrics used in roofing	ASTM D5665; D5726

R905.9.2.1

Rosin paper shall be used when the membrane is applied directly to a wood deck or cementitious fiber decks.

R905.9.3 Application

Reserved.

R905.10 Metal Roof Panels

The installation of metal roof panels shall comply with the provisions of this section. Metal roofing panels shall be factory or field manufactured in accordance with the manufacturers' product approval specifications and limitations of use. Metal roofing panels shall be factory or field manufactured under a quality assurance program that is audited by a third-party quality assurance entity approved by the Florida Building Commission for that purpose.

R905.10.1 Deck Requirements

Metal roof panel roof coverings shall be applied to solid or spaced sheathing, except where the roof covering is specifically designed to be applied to spaced supports.

R905.10.2 Slope

Minimum slopes for metal roof panels shall comply with the following:

1. The minimum slope for lapped, nonsoldered-seam metal roofs without applied lap sealant shall be three units vertical in 12 units horizontal (25-percent slope).
2. The minimum slope for lapped, nonsoldered-seam metal roofs with applied lap sealant shall be one-half unit vertical in 12 units horizontal (4-percent slope). Lap sealants shall be applied in accordance with the *approved* manufacturer's installation instructions.
3. The minimum slope for standing-seam roof systems shall be one-quarter unit vertical in 12 units horizontal (2-percent slope).

R905.10.3 Material Standards

Metal-sheet roof covering systems that incorporate supporting structural members shall be designed in accordance with the *Florida Building Code, Building*. Metal-sheet roof coverings installed over structural decking shall comply with Table R905.4.4. The materials used for metal-sheet roof coverings shall be naturally corrosion resistant or provided with corrosion resistance in accordance with the standards and minimum thicknesses shown in Table R905.4.4.

Table R905.10.3(1) Metal Roof Covering Standards. Reserved.

Table R905.10.3(2) Minimum Corrosion Resistance. Reserved.

R905.10.4 Attachment

Metal roof panels shall be secured to the supports in accordance with this chapter and the manufacturer's installation instructions. In the absence of manufacturer's installation instructions, the following fasteners shall be used:

1. Galvanized fasteners shall be used for steel roofs.
2. Copper, brass, bronze, copper alloy and 300-series stainless steel fasteners shall be used for copper roofs.
3. Stainless steel fasteners are acceptable for metal roofs.
4. Aluminum-zinc coated fasteners are acceptable for aluminum-zinc coated roofs.

R905.10.5 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.11.

R905.11 Modified Bitumen Roofing

The installation of modified bitumen roofing shall comply with the provisions of this section.

R905.11.1 Slope

Modified bitumen membrane roofs shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) for drainage.

R905.11.2 Material Standards

Modified bitumen roof coverings shall comply with the standards in Table R905.11.2.

TABLE R905.11.2

MODIFIED BITUMEN ROOFING MATERIAL STANDARDS

MATERIAL	STANDARD
Acrylic coating	ASTM D6083
Asphalt adhesive	ASTM D3747

Asphalt cement	ASTM D3019
Asphalt coating	ASTM D1227; D2824
Asphalt primer	ASTM D41
Modified bitumen roof membrane	ASTM D6162; D6163; D6164; D6222; D6223; D6298; D6509

R905.11.3 Application

Modified bitumen roofs shall be installed in accordance with this chapter and the manufacturer's instructions. The approved allowable uplift resistance for the modified bitumen roof shall be equal to or greater than the uplift resistance for the roof based on Table R301.2(2).

R905.12 Thermoset Single-Ply Roofing

The installation of thermoset single-ply roofing shall comply with the provisions of this section.

R905.12.1 Slope

Thermoset single-ply membrane roofs shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) for drainage.

R905.12.2 Material Standards

Thermoset single-ply roof coverings shall comply with ASTM D4637 or ASTM D5019.

R905.12.3 Application

Thermoset single-ply roofs shall be installed in accordance with this chapter and the manufacturer's instructions. The approved allowable uplift resistance for the thermoset single-ply membrane roof shall be equal to or greater than the uplift resistance for the roof based on Table R301.2(2).

R905.13 Thermoplastic Single-Ply Roofing

The installation of thermoplastic single-ply roofing shall comply with the provisions of this section.

R905.13.1 Slope

Thermoplastic single-ply membrane roofs shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope).

R905.13.2 Material Standards

Thermoplastic single-ply roof coverings shall comply with ASTM D4434, ASTM D6754 or ASTM D6878.

R905.13.3 Application

Thermoplastic single-ply roofs shall be installed in accordance with this chapter and the manufacturer's instructions. The approved allowable uplift resistance for the thermoplastic single-ply roof shall be equal to or greater than the uplift resistance for the roof based on Table R301.2(2).

R905.14 Sprayed Polyurethane Foam Roofing

The installation of sprayed polyurethane foam roofing shall comply with the provisions of this section or in compliance with RAS 109 and 109A.

R905.14.1 Slope

Sprayed polyurethane foam roofs shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) for drainage.

R905.14.2 Material Standards

Spray-applied polyurethane foam insulation shall comply with ASTM C1029, Type III or IV or ASTM D7425.

R905.14.3 Application

Foamed-in-place roof insulation shall be installed in accordance with this chapter and the manufacturer's instructions. A liquid-applied protective coating that complies with Table R905.14.3 shall be applied not less than 2 hours nor more than 72 hours following the application of the foam. The approved allowable uplift resistance for the sprayed polyurethane foam roofing shall be equal to or greater than the uplift resistance for the roof based on Table R301.2(2).

TABLE R905.14.3
PROTECTIVE COATING MATERIAL STANDARDS

MATERIAL	STANDARD
Acrylic coating	ASTM D6083
Silicone coating	ASTM D6694
Moisture-cured polyurethane coating	ASTM D6947

R905.14.4 Foam Plastics

Foam plastic materials and installation shall comply with Section R316.

R905.15 Liquid-Applied Roofing

The installation of liquid-applied roofing shall comply with the provisions of this section.

R905.15.1 Slope

Liquid-applied roofing shall have a design slope of not less than one-fourth unit vertical in 12 units horizontal (2-percent slope).

R905.15.2 Material Standards

Liquid-applied roofing shall comply with ASTM C836, C957, D1227, D3468, D6083, D6694 or D6947.

R905.15.3 Application

Liquid-applied roofing shall be installed in accordance with this chapter and the manufacturer's approved installation instructions. The approved allowable uplift resistance for the liquid-applied coatings shall be equal to or greater than the uplift resistance for the roof based on Table R301.2(2).

R905.16 Building-Integrated Photovoltaic Roofing Modules/Shingles

The installation of building-integrated photovoltaic roofing modules/shingles shall comply with the provisions of this section, Section R324 and NFPA 70.

R905.16.1 Deck Requirements

Reserved.

R905.16.2 Deck Slope

Reserved.

R905.16.3 Underlayment

Underlayment shall comply and be installed in accordance with Section R905.11.

R905.16.4 Underlayment Application

Reserved.

R905.16.4.1 Ice Barrier

Reserved.

R905.16.4.2 Underlayment and High Winds

Reserved.

R905.16.5 Material Standards

Building-integrated photovoltaic roofing modules/shingles shall be listed and labeled in accordance with UL 1703.

R905.16.6 Attachment

Building-integrated photovoltaic roofing modules/shingles shall be attached in accordance with the manufacturer's installation instructions.

R905.16.7 Wind Resistance

Building-integrated photovoltaic roofing modules/shingles shall be tested in accordance with procedures and acceptance criteria in ASTM D3161 or TAS 107. Building-integrated photovoltaic roofing modules/shingles shall comply with the classification requirements of Table R905.2.6.1 for the appropriate maximum basic wind speed. Building-integrated photovoltaic roofing modules/shingle packaging shall bear a label to indicate compliance with the procedures in ASTM D3161 or TAS 107 and the required classification from Table R905.2.6.1.

R905.17 Photovoltaic Systems

Rooftop mounted photovoltaic systems shall be designed in accordance with this section.

R905.17.1 Wind Resistance

Rooftop mounted photovoltaic systems shall be designed for wind loads in accordance with ASCE 7.

R905.17.2 Fire Classification

Rooftop mounted photovoltaic systems shall have the same fire classification as required for the roof assembly by Section R902.

R905.17.3 Installation

Rooftop mounted photovoltaic systems shall be installed in accordance with the manufacturer's installation instructions.

R905.17.4 Photovoltaic Panels and Modules

Photovoltaic panels and modules mounted on top of a roof shall be listed and labeled in accordance with UL 1703 and shall be installed in accordance with the manufacturer's installation instructions.

Section R906 Roof Insulation

R906.1 General

Where above-deck thermal insulation is installed, such insulation shall be covered with an *approved* roof covering and shall comply with NFPA 276 or UL 1256.

R906.2 Material Standards

Above-deck thermal insulation board shall comply with the standards in Table R906.2.

TABLE R906.2

MATERIAL STANDARDS FOR ROOF INSULATION

Cellular glass board	ASTM C552 or ASTM C1902
Mineral wool board	ASTM C726
Composite boards	ASTM C1289, Type III, IV, V or VI
Expanded polystyrene	ASTM C578
Extruded polystyrene board	ASTM C578
Perlite board	ASTM C728

Polyisocyanurate board	ASTM C1289, Type I or II
Wood fiberboard	ASTM C208
Fiber-reinforced gypsum board	ASTM C1278
Glass-faced gypsum board	ASTM C1177

Section R907 Rooftop-Mounted Photovoltaic Systems

R907.1 Rooftop-Mounted Photovoltaic Systems

Rooftop-mounted photovoltaic panel systems shall be designed and installed in accordance with Section R324, NFPA 70 and the *Florida Fire Prevention Code*.

Section R908 Existing Roofing

R908.1 General

Materials and methods of application used for re-covering or replacing an existing roof covering shall comply with the requirements of this chapter.

Exceptions:

1. Reroofing shall not be required to meet the minimum design slope requirement of one-quarter unit vertical in 12 units horizontal (2-percent slope) in Section R905 for roofs that provide positive roof drainage.
2. An existing self-adhering modified bitumen underlayment that has been previously installed over the roof decking and where it is required, reroofing of the roof sheathing in accordance with Section R908.7.1 can be confirmed or verified. An approved underlayment in accordance with Table R905.1.1.1 for the applicable roof covering shall be applied over the entire roof over the existing self-adhered modified bitumen underlayment.

R908.1.1

Not more than 25 percent of the total roof area or roof section of any existing building or structure shall be repaired, replaced or recovered in any 12-month period unless the roof covering on the entire existing roof system or roof section is replaced or recovered to conform to the requirements of this code.

Exception: If an existing roofing system or roof section was built, repaired, or replaced in compliance with the requirements of the 2007 *Florida Building Code*, or any subsequent editions of the *Florida Building Code*, and 25 percent or more of such roofing system or roof section is being repaired, replaced, or recovered, only the repaired, replaced, or recovered portion is required to be constructed in accordance with the *Florida Building Code* in effect, as applicable. Pursuant to s. 553.844(5), *Florida Statutes*, a local government may not adopt by ordinance an administrative or technical amendment to this exception.

R908.2 Structural and Construction Loads

The structural roof components shall be capable of supporting the roof covering system and the material and equipment loads that will be encountered during installation of the roof covering system.

R908.3 Recovering Versus Replacement

New roof coverings shall not be installed without first removing all existing layers of roof coverings where any of the following conditions occur:

1. Where the existing roof or roof covering is water soaked or has deteriorated to the point that the existing roof or roof covering is not adequate as a base for additional roofing.
2. Where the existing roof covering is wood shingle or shake, slate, clay, cement or asbestos-cement tile.

3. Where the existing roof has two or more applications of any type of roof covering.
4. When blisters exist in any roofing, unless blisters are cut or scraped open and remaining materials secured down before applying additional roofing.
5. Where the existing roof covering is to be used for attachment for a new roof system and compliance with the securement provisions of Section R905 cannot be met.

Exceptions:

1. Complete and separate roofing systems, such as standing-seam metal roof systems, that are designed to transmit the roof loads directly to the building's structural system and that do not rely on existing roofs and roof coverings for support, shall not require the removal of existing roof coverings.
2. Reserved.
3. The application of new protective coating over existing spray polyurethane foam roofing systems shall be permitted without tear-off of existing roof coverings.
4. Reserved.
5. Roof coating. Application of elastomeric and/or maintenance coating systems over existing asphalt shingles shall be in accordance with the shingle manufacturer's approved installation instructions.

R908.3.1 Roof Re-cover

Reserved.

R908.3.1.1 Roof Re-cover Not Allowed

Reserved.

R908.4 Roof Re-covering

Reserved.

R908.5 Reinstallation/Reuse of Materials

Existing or salvaged slate, clay or concrete tile shall be permitted for reinstallation or reuse, to repair an existing slate or tile roof, except that salvaged slate or tile shall be of like kind in both material and profile. Damaged, cracked or broken slate or tile shall not be reinstalled. The building official may permit salvaged slate, clay and concrete tile to be installed on additions and new construction, when the tile is tested in compliance with the provisions of Section R905 or Section 1523 of the *Florida Building Code, Building* (HVHZ shall comply with Section 1523 of the *Florida Building Code, Building*) and installed in accordance with Section R905 or Section 1518 of the *Florida Building Code, Building* (HVHZ shall comply with Section 1518 of the *Florida Building Code, Building*). Existing vent flashing, metal edgings, drain outlets, collars and metal counterflashings shall not be reinstalled where rusted, damaged or deteriorated. Aggregate surfacing materials shall not be reinstalled (High-Velocity Hurricane Zones shall comply with Sections 1512 through 1525 of the *Florida Building Code, Building*).

R908.6 Flashings

Flashings shall be reconstructed in accordance with *approved* manufacturer's installation instructions or in compliance with RAS 111. Metal flashing to which bituminous materials are to be adhered shall be primed prior to installation.

R908.7 Wind Mitigation

When a roof covering on an existing structure with a sawn lumber, wood plank or wood structural panel roof deck is removed and replaced, the following procedures shall be permitted to be performed by the roofing contractor:

- (a) Roof-decking attachment shall be as required by Section R908.7.1.

- (b) A secondary water barrier shall be provided as required by Section R908.7.2.

Exception: Structures permitted subject to the *Florida Building Code* are not required to comply with this section.

R908.7.1 Roof Decking Attachment for Existing Structures With Wood Roof Decks

Fastening for sawn lumber, wood plank or wood structural panel roof decks shall be in accordance with Section R908.7.1.1 or R908.7.1.2 as appropriate for the existing construction. 8d nails shall be a minimum of 0.113 inch (2.9 mm) in diameter and shall be a minimum of 2¹/₄ inch (57 mm) long to qualify for the provisions of this section for existing nails regardless of head shape or head diameter.

R908.7.1.1

Roof decking consisting of sawn lumber or wood planks up to 12 inches wide and secured with at least two nails (minimum size 8d) to each roof framing member it crosses shall be deemed to be sufficiently connected. Sawn lumber or wood plank decking secured with smaller fasteners than 8d nails or with fewer than two nails (minimum size 8d) to each framing member it crosses shall be deemed sufficiently connected if fasteners are added such that two clipped head, round head, or ring shank nails (minimum size 8d) are in place on each framing member it crosses.

R908.7.1.2

For roof decking consisting of wood structural panels, fasteners and spacing required in columns 3 and 4 of Table R908.7.1.2 are deemed to comply with the indicated design wind speed range. Wood structural panel connections retrofitted with a two part urethane based closed cell adhesive sprayed onto the joint between the sheathing and framing members are deemed to comply provided testing using the manufacturer's recommended application on panels connected with 6d smooth shank nails at no more than a 6-inch edge and 12-inch field spacing demonstrate an uplift resistance of a minimum of 200 psf.

Supplemental fasteners as required by Table R908.7.1.2 shall be ASTM F1667 classification RSRS-01 ring shank nails with the following minimum dimensions:

1. 0.113-inch nominal shank diameter.
2. Ring diameter a minimum of 0.012-inch greater than shank diameter.
3. 16 to 20 rings per inch.
4. A minimum 0.280-inch full round head diameter.
5. Ring shank to extend a minimum of 1¹/₂ inches from the tip of the nail.
6. Minimum 2¹/₄ inch nail length.

TABLE R908.7.1.2

SUPPLEMENT FASTENERS AT PANEL EDGES AND INTERMEDIATE FRAMING

EXISTING FASTENERS	EXISTING SPACING	V_{asd} 110 MPH OR LESS SUPPLEMENTAL FASTENER SPACING SHALL BE NO GREATER THAN	V_{asd} GREATER THAN 110 MPH SUPPLEMENTAL FASTENER SPACING SHALL BE NO GREATER THAN
Staples or 6d	Any	6 inches o.c. ^b	6 inches o.c. ^b

8d clipped head, round head, smooth or ring shank	6 inches o.c. or less	None necessary	None necessary
8d clipped head, round head, smooth or ring shank	Greater than 6 inches o.c.	6 inches o.c. ^a	6 inches o.c. ^a

For SI: 1 inch = 25.4 mm.

- a. Maximum spacing determined based on existing fasteners and supplemental fasteners.
- b. Maximum spacing determined based on supplemental fasteners only.
- c. V_{asd} shall be determined in accordance with Section 1609.3.1 of the *Florida Building Code, Building* or Section R301.2.1.3 of the *Florida Building Code, Residential*.

R908.7.2 Roof Secondary Water Barrier for Existing Structures With Wood Roof Decks

When a roof covering is removed and replaced, a secondary water barrier shall be installed in accordance with Section R905.1.1 or Section 1518.2 of the *Florida Building Code, Building*.

Exceptions:

- 1. Roof slopes < 2:12 having a continuous roof system shall be deemed to comply with Section R908.7.2 requirements for a secondary water barrier.
- 2. Clay and concrete tile roof systems installed as required by the *Florida Building Code, Residential* are deemed to comply with the requirements of Section R908.7.2 for Secondary Water Barriers.

R908.8

When a roof covering on an existing structure with a sawn lumber, wood plank or wood structural panel roof deck is removed and replaced on a building that is located in the wind-borne debris region as defined in the *Florida Building Code, Building* and that has an insured value of \$300,000 or more or, if the building is uninsured or for which documentation of insured value is not presented, has a just valuation for the structure for purposes of ad valorem taxation of \$300,000 or more:

- (a) Roof to wall connections shall be improved as required by Section R908.8.1.
- (b) Mandated retrofits of the roof-to-wall connection shall not be required beyond a 15 percent increase in the cost of re-roofing.

Exception: Structures permitted subject to the *Florida Building Code* are not required to comply with this section.

R908.8.1 Roof-to-Wall Connections for Structures With Wood Roof Decks

Where required by Section R908.8, the intersection of roof framing with the wall below shall provide sufficient resistance to meet the uplift loads specified in Table R908.8.1 either because of existing conditions or through retrofit measures. As an alternative to an engineered design, the prescriptive retrofit solutions provided in Sections R908.8.1.1 through R908.8.1.7 shall be accepted as meeting the mandated roof-to-wall retrofit requirements.

Exceptions:

1. Where it can be demonstrated (by code adoption date documentation and permit issuance date) that roof-to-wall connections and/or roof-to-foundation continuous load path requirements were required at the time of original construction.
2. Roof-to-wall connections shall not be required unless evaluation and installation of connections at gable ends or all corners can be completed for 15 percent of the cost of roof replacement.

TABLE R908.8.1

REQUIRED UPLIFT CAPACITIES FOR ROOF-TO-WALL CONNECTIONS^{a, b, c} (POUNDS PER LINEAL FOOT)

	Ultimate Design Wind Speed, V_{ult}^d	Roof Span (feet)							Overhangs
		12	20	24	28	32	36	40	
Within 6 feet of building corner	85	-69.85	-116.42	-139.70	-162.99	-186.27	-209.55	-232.84	-27
	90	-82.67	-137.78	-165.34	-192.90	-220.45	-248.01	-275.57	-30.3
	100	-110.51	-184.18	-221.01	-257.85	-294.68	-331.52	-368.36	-37.4
	110	-141.27	-235.45	-282.55	-329.64	-376.73	-423.82	-470.91	-45.3
	120	-174.97	-291.62	-349.94	-408.26	-466.59	-524.91	-583.23	-53.9
	130	-211.60	-352.66	-423.19	-493.72	-564.26	-634.79	-705.32	-63.2
	140	-251.15	-418.59	-502.31	-586.02	-669.74	-753.46	-837.18	-73.3
	150	-293.64	-489.40	-587.28	-685.16	-783.04	-880.92	-978.80	-84.2
	170	-387.40	-645.67	-774.81	-903.94	-1033.08	-1162.21	-1291.35	-108
Greater than 6 feet from building corner	85	-39.10	-65.17	-78.20	-91.24	-104.27	-117.30	-130.34	-27
	90	-48.20	-80.33	-96.39	-112.46	-128.52	-144.59	-160.66	-30.3
	100	-67.95	-113.24	-135.89	-158.54	-181.19	-203.84	-226.49	-37.4
	110	-89.78	-149.63	-179.55	-209.48	-239.40	-269.33	-299.25	-45.3
	120	-113.68	-189.47	-227.37	-265.26	-303.16	-341.05	-378.94	-53.9
	130	-139.67	-232.78	-279.34	-325.90	-372.45	-419.01	-465.57	-63.2
	140	-167.74	-279.56	-335.47	-391.38	-447.29	-503.21	-559.12	-73.3
	150	-197.88	-329.80	-395.76	-461.72	-527.68	-593.64	-659.60	-84.2
	170	-264.41	-440.68	-528.81	-616.95	-705.08	-793.22	-881.35	-108

For SI: 1 foot = 304.8 mm; 1 pound per lineal foot = 1.488 kg/m; 1 mile per hour = 0.305 m/s.

- a. The uplift loads are pounds per lineal foot of building length. For roof uplift connections multiply by 1.33 for framing spaced 16 inches on center and multiply by 2 for framing spaced 24 inches on center.
- b. The uplift loads do not account for the effects of overhangs. The magnitude of the above loads shall be increased by adding the overhang loads found in the table. The overhang loads are also based on framing spaced 12 inches

on center. The overhang loads given shall be multiplied by the overhang projection and added to the roof uplift value in the table.

- c. For Ultimate Design Wind Speeds, V_{ult} , greater than 170 mph, wind uplift forces shall be determined in accordance with *Florida Building Code, Residential* Section R301.2.1.1 or ASCE 7.
- d. Ultimate Design Wind Speeds determined from Figure 1609.3(1) in the *Florida Building Code, Building* or Figure R301.2(4) in the *Florida Building Code, Residential*.

R908.8.1.1 Access for Retrofitting Roof to Wall Connections

These provisions are not intended to limit the means for gaining access to the structural elements of the roof and wall for the purposes of retrofitting the connection. The retrofit of roof to wall connections can be made by access through the area under the eave, from above through the roof, or from the interior of the house. Methods for above access include removal of roof panels or sections thereof or removal of portions of roof paneling at selected locations large enough for access, viewing, and installing the retrofit connectors and fasteners.

Where panels or sections are removed, the removed portions shall not be reused. New paneling shall be used and fastened as in new construction.

Holes shall be deemed adequately repaired if a patch of paneling is installed with no gap greater than $\frac{1}{2}$ inch (13 mm) between the patch and the existing sheathing and if the patch is supported using one of the following methods.

- a) Solid $\frac{1}{2}$ -inch lumber shall fully support the patch and shall be secured to the existing sheathing with #8 by $\frac{1}{4}$ -inch screws spaced a minimum of 3 inches (76 mm) around the perimeter with screws a minimum of $\frac{3}{4}$ inch from the near edge of the hole. The patch shall be secured to the lumber with #8 $\times$ $\frac{1}{4}$ -inch-screws spaced on a grid no greater than 6 inches by 6 inches (152 mm $\times$ 152 mm) with no fewer than 2 screws.
- b) Holes that extend horizontally from roof framing member to adjacent roofing framing member that are less than or equal to 7 inches (178 mm) wide along the slope of the roof shall be supported by a minimum of 2 $\times$ 4 lumber whose face is attached to each roofing framing members using a minimum of 2 each 3-inch (76 mm) long fasteners (#8 screws or 10d common nails) connecting the two. The patch shall have attached to its bottom, running horizontally, a minimum 2 $\times$ 4 either flat wise or on edge secured with #8 $\times$ $\frac{1}{4}$ inch screws a maximum of 4 inches (102 mm) on center and no more distant from the end of the added lumber than 3 inches (76 mm). The patch shall be secured with two #8 $\times$ 1- $\frac{1}{4}$ -inch screws to each support member.

R908.8.1.2 Partially Inaccessible Straps

Where part of a strap is inaccessible, if the portion of the strap that is observed is fastened in compliance with these requirements, the inaccessible portion of the strap shall be presumed to comply with these requirements.

R908.8.1.3 Prescriptive Method for Gable Roofs on a Wood Frame Wall

The anchorage of each of the exposed rafters or truss within 6 feet (1829 mm) of the corner along the exterior wall on each side of each gable end shall be inspected. Wherever a strap is missing or an existing strap has fewer than four fasteners on each end, approved straps, ties or right angle brackets with a minimum uplift capacity of 500 pounds (740 kg) shall be installed that connect each rafter or truss to the top plate below. Adding fasteners to existing straps shall be allowed in lieu of adding a new strap provided the strap is manufactured to accommodate at least 4 fasteners at each end. Wherever access makes it possible (without damage of the wall or soffit finishes), both top plate members shall be connected to the stud below using a stud to plate connector with a minimum uplift capacity of 500 pounds (740 kg). Use of straps that connect directly from the rafter or truss to the wall stud below shall be allowed as an alternate provided the two members align with no more than $\frac{1}{2}$ inches (38 mm) offset.

R908.8.1.4 Prescriptive Method for Gable Roofs on a Masonry Wall

The anchorage of each of the exposed rafters or trusses within 6 feet (1829 mm) of the corner along the exterior wall on each side of each gable end shall be inspected. Wherever a strap is missing or an existing strap has fewer than four fasteners on each end, approved straps, ties or right angle gusset brackets with a minimum uplift capacity of 500 pounds (740 kg) shall be

installed that connect each rafter or truss to the top plate below or directly to the masonry wall using approved masonry screws of a length and diameter recommended by the manufacturer. In the absence of manufacturer's recommendations, screws shall provide at least a 2½-inch (64 mm) embedment into the concrete or masonry. When the straps or right angle gusset brackets are attached to a wood sill plate, the sill plate shall be anchored to the concrete masonry wall below. This anchorage shall be accomplished by installing ¼-inch diameter masonry screws, each with supplementary ¼-inch washer, having sufficient length to develop a 2½-inch (64 mm) embedment into the concrete and masonry. These screws shall be installed within 4 inches (102 mm) of the truss or rafter on both sides of each interior rafter or truss and on the accessible wall side of the gable end truss or rafter.

R908.8.1.5 Prescriptive Method for Hip Roofs on a Wood Frame Wall

Unless it is possible to verify through nondestructive inspection or from plans prepared by a design professional that the roof structure is anchored at least as well as outlined below, access shall be provided at a minimum to the hip rafter (commonly known as a "king jack"), to the hip girder and at each corner of the hip roof. The hip rafter (commonly known as a "king jack"), the hip girder and the rafters/trusses adjacent to the hip girder that are not anchored with a strap having at least four fasteners on each end, shall be connected to the top plate below using a strap or a right angle gusset bracket having a minimum uplift capacity of 500 pounds (740 kg). Adding fasteners to existing straps shall be allowed in lieu of adding a new strap provided the strap is manufactured to accommodate at least four fasteners at each end. Wherever access makes it possible (without damage of the wall or soffit finishes), both top plate members shall be connected to the stud below using a stud to plate connector with a minimum uplift capacity of 500 pounds (740 kg). Use of straps that connect directly from the hip rafter, hip girder or adjacent rafters/trusses to the wall stud below shall be allowed as an alternate provided the two members align with no more than 1½ inch (38 mm) offset.

R908.8.1.6 Prescriptive Method for Hip Roofs on a Masonry Wall

Unless it is possible to verify through nondestructive inspection or from plans prepared by a design professional that the roof structure is anchored at least as well as outlined below, access shall be provided at a minimum to the hip rafter (commonly known as a "king jack"), to the hip girder and at each corner of the hip roof. The hip rafter (commonly known as a "king jack"), the hip girder and the rafters/trusses adjacent to the hip girder that are not anchored with a strap having at least four fasteners on each end, shall be connected to the concrete masonry wall below using approved straps or right angle gusset brackets with a minimum uplift capacity of 500 pounds (740 kg). Adding fasteners to existing straps shall be allowed in lieu of adding a new strap provided the strap is manufactured to accommodate at least four fasteners at each end. The straps or right angle gusset brackets shall be installed such that they connect each rafter or truss to the top plate below or directly to the masonry wall using approved masonry screws of a length and diameter recommended by the manufacturer. In the absence of manufacturer's recommendations, screws shall provide at least 2½-inches (64 mm) embedment into the concrete or masonry. When the straps or right angle gusset brackets are attached to a wood sill plate, the sill plate shall be anchored to the concrete masonry wall below. This anchorage shall be accomplished by installing ¼-inch (6 mm) diameter masonry screws, each with supplementary ¼-inch (6 mm) washer, with sufficient length to develop a 2½-inch (64 mm) embedment into the concrete and masonry. These screws shall be installed within 4 inches (102 mm) of the truss or rafter on both sides of each interior rafter or truss and on the accessible wall side of the gable end truss or rafter.

R908.8.1.7 Priorities for Mandated Roof-to-Wall Retrofit Expenditures

Priority shall be given to connecting the exterior corners of roofs to walls where the spans of the roofing members are greatest. For houses with both hip and gable roof ends, the priority shall be to retrofit the gable end roof-to-wall connections unless the width of the hip end is more than 1.5 times greater than the width of the gable end. When considering priorities for houses with both hip and gable roof ends, and the 15 percent of the cost of roof replacement is sufficient to complete all of the prioritized elements pursuant to this section, but is not sufficient to complete all of the non-prioritized elements, then no portion of complete retrofit of the non-prioritized element is required.